

Abdul Khurram

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Available for Co-ops/Internships (Open to Relocation): May-Dec 2026 | Full-time: May 2027+

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Python, Scala, Kafka, Airflow, Docker, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered **distributed Python/Scala** services processing **50M+ daily transactions** via **Kafka** streams and **Airflow DAGs**, enforcing strict **RBAC/encryption** on **petabyte-scale** financial data.
- Optimized **high-throughput** batch jobs by migrating to **Spark-native APIs** and tuning **AQE/cluster memory**, reducing **latency by 40%** and resolving **resource contention** bottlenecks.
- Shipping **multi-region microservices** with **fault tolerance**, **idempotent** re-runs, and automated **failover** for **high-reliability** banking workloads.

Software Engineer Intern | *Python/Flask, TypeScript/Node.js, AWS, Docker*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Built **distributed async** backend services on **AWS (SQS/S3)** with **Python/Flask REST APIs**, processing **thousands of requests/day** with **fault-tolerant** message passing and stream-based automation.
- Raised **service reliability** to **99.9%** via **retry/backoff**, **Docker**-containerized deployments, **E2E test** gates in **CI/CD**, and **CloudWatch** monitoring/alerting.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led development of an **AI-driven** VR clinical simulation in **Unreal (C++)**, collaborating with UBC's School of Nursing to train nurse practitioners.
- Designed a novel **multithreaded supervisory architecture (C++/Python)** that monitors AI patient behavior and enforces clinical protocol adherence.
- Evaluated system performance, achieving **40%** latency reduction and **90%** inference cost savings through pipeline optimizations.

PROJECTS

Backer | *TypeScript/Next.js, AWS (DynamoDB, S3, SES), NextAuth*

Hack The Coast 2026

- Architected a **DynamoDB**-backed data layer (**10+ tables**) with typed CRUD, batch operations, and **transactional writes** powering an investor feed, messaging, and investment workflows.
- Trained an **XGBoost ranker** scoring startups on 6 features (stage affinity, text similarity, tag overlap, popularity) from **DynamoDB** event streams.
- Integrated **S3** uploads, **SES** emails, and **NextAuth** middleware with **role-based protection**.

Workbase | *TypeScript/Next.js, Prisma/PostgreSQL, AWS Bedrock, Vitest*

Personal Project

- Built a full-stack platform with **Prisma/PostgreSQL** data layer, **REST API** routes, and **GitHub OAuth** service integration.
- Designed a multi-step processing pipeline via **AWS Bedrock** with **Zod** schema validation, structured output repair, and generation run auditing.

SKILLS

Languages: Python, Java, Scala, SQL, TypeScript/JavaScript, Go, C/C++

Platforms & Tools: AWS (SQS, S3, Lambda, DynamoDB), **Kafka**, **Docker**, **PostgreSQL**, **Redis**; **REST APIs**, **Flask/FastAPI**, **Airflow**, **Git**, **Linux**